

2026 Spring, CMPE 442 Introduction to Machine Learning

Programming Assignment-3

In this assignment, you are expected to implement K-Means and Gaussian Mixture Models (GMM) for **image segmentation**. You are given 4 different images (taken from COCO 2017 validation set) on LMS to experiment with. Read the following instructions carefully before starting to work on the assignment.

- You are expected to implement K-Means and GMM from scratch. You are not allowed to use a ready-to-use implementation of a machine learning library (scikit-learn, etc.). However, you are allowed to use any libraries (numpy, scipy, pandas, matplotlib, scikit-image, etc.) for matrix operations, visualization, data manipulation, mathematical operations, image operations, etc.
- You are expected to report your results using 4 different values for the number of clusters (K). Since the initialization affects the outcome, it is recommended to run the algorithms multiple times for each K and pick the best result to report.
- To prevent any problems with folder locations, make sure that the provided images folder is located in the same folder with your code files. Use relative paths instead of absolute paths (Check the [link](#) for a comparison of relative paths and absolute paths).
- You must prepare a report that contains the following:
 - o The original images and segmentation maps produced by both algorithms
(4 images x 4 different values for K x 2 algorithms = 32 segmentation maps in total)
 - o A brief comparative analysis of segmentation maps produced by the algorithms.
- The report must not exceed 1 page excluding the visuals.
- You have to use Python programming language for this assignment.
- It is advised to add comments to your codes.
- You are allowed to choose the values of hyperparameters.
- The provided VPL will not provide any test case evaluation.